

Inclusive Innovation for U.S. Economic Growth and Resiliency

By Tom Guevara, Alexander Kersten, and Srishti Khemka

*In December 2022, the Renewing American Innovation Project at CSIS hosted a webinar titled “**Inclusive Innovation: Challenges and Paths Forward**.” The event, convened in cooperation with the Indiana University Public Policy Institute, brought together professionals from academia, policy, and industry to assess the challenges of inclusion, as well as to discuss the promise of the U.S. patent system as an important tool for empowering diverse and underrepresented inventors.*

Introduction

Innovation—the process of turning ideas into products and services for the marketplace—is the engine that powers the knowledge economy. Entrepreneurs drive this innovation, with the late Austrian American management expert Peter Drucker **arguing** that “innovation is the specific function of entrepreneurship, whether in an existing business, a public service institution, or a new venture started by a lone individual in the family kitchen.”

Broadening the ranks of entrepreneurs can open new opportunities and allow for the growth of more inclusive and collaborative networks across research universities and laboratories. It can also foster more inclusive participation in the knowledge economy across the nation’s demographics and regions. Moreover, when Americans from all backgrounds see the innovation system as a common pool resource that works for them, they are in turn more invested in the energy that powers the nation’s economic growth and enhances its competitiveness.

How Inclusion Improves Innovation

Indeed, it has been shown that greater inclusion in the entrepreneurship and innovation ecosystems significantly improves economic growth. A 2020 Citigroup **report** estimated that including more

women and Black Americans in the initial stages of innovation could increase U.S. GDP by as much as \$640 billion. The report further argued that the United States' aggregate economic output would be \$16 trillion higher today if identified racial gaps had been closed in 2000, which would be almost double that year's GDP.

Correspondingly, a new study argues that disenfranchising and failing to support inclusion efforts harms the nation's economy. According to a 2023 McKinsey **report**, *Rekindling US productivity for a new era*, U.S. productivity growth—essential for increasing incomes, mitigating inflation, and increasing employment—has faltered for the past 15 years, in part due to “the unevenness with which the benefits of innovation, globalization, and other shapers of productivity have spread” across the country.

Greater inclusion also assists the United States in renewing its innovation economy. U.S. patent experts and leading industry experts recognize the importance of increasing diversity in the innovation ecosystem not only for its economic benefits, but also because of how it improves the innovation ecosystem itself. At the **December 2022 CSIS event**, Jackson State University assistant vice president for research and economic development Almesha Campbell noted that “inclusive cultures are six times more likely to be innovative.” She noted that inclusivity increases opportunities for collaboration among all participating members of an organization or ecosystem, resulting in a greater number of novel ideas generated and greater success in their commercialization.

Despite the logic of this evidence, most businesses in the high-tech sector have **struggled** to recruit diverse talent at every level of employment. This sector has historically been and continues to be more white and male-dominated compared to the overall private industry, according to a **report** from the United States Equal Employment Opportunity Commission.

A Shift toward Greater Diversity

Seeking to change this state of affairs, technology firms have sought to intensify their diversity outreach efforts in recent years, **spending** more than \$8 billion annually on diversity, equity, inclusion (DEI) training and improved recruitment efforts. For example, in its 2021 **annual diversity report** titled *We're listening, learning, and taking action.*, Google admitted its workforce could be far more representative of the diversity of the nation and announced efforts to tackle the issue head on by “working with external partner organizations to build community and provide resources to groups historically excluded from the tech industry.”

The ongoing shift in work habits, accelerated by the Covid-19 pandemic, may also be altering the demographics of the U.S. innovator. Analysis from CSIS in 2021 **found** that the decades-long pattern of venture capital firms investing only in their own coastal, urban environments might finally be **changing** as the work-from-anywhere phenomenon takes hold. Already in late 2020, it had become clear that the pandemic was perhaps prompting “creative destruction,” in which applications for new businesses in the United States were **rising** at their fastest rate since 2007.

Female entrepreneurship has also grown over the past few years. The Global Entrepreneurship Monitor's (GEM) report ***Women's Entrepreneurship 2020/21: Thriving Through Crisis*** documents a rise in entrepreneurship among Black and Hispanic female entrepreneurs linked to increases to support from the Economic Development Administration (EDA) within the U.S. Department of Commerce. In October

2022, the Commerce Department **announced** its inclusion initiative of “Build to Scale” grants totaling \$47 million, with the goal of driving innovation and tech-based economic development.

Understanding Current Structural Rigidities

More diverse participation can also be an effective tool for breaking down the existing structural rigidities that remain in the workforce. The Kauffman Foundation’s *Who is the Entrepreneur? New Entrepreneurs in the United States, 1996-2021* report notes that even though there has been a great amount of female entrepreneurship over the course of decades in the United States, the share of women among new entrepreneurs has remained relatively consistent over the past 25 years. The authors found that the pandemic’s disruptions have not proven to be as transformational as it appeared they would be in late 2020, citing the impact of the pandemic on the global financial environment—one that forced **massive layoffs** in the tech industry and made **venture capital** wary of expanding its investments in recent months. The banking crisis that began with the **collapse** of Silicon Valley Bank in March 2023 has likely only worsened this problem, especially as relatively high interest rates appear to be ushering in a new era for consumer technology firms dependent on venture funding.

Also of concern is data from the Congressional Budget Office (CBO) showing that overall entrepreneurship has been in **decline** in the United States since the late 1970s, with the percentage of young businesses compared to all businesses within the economy having fallen by half. However, GEM has **claimed** that the CBO dataset may not be capturing the full story because it looks narrowly at government census data. Based on its polling, GEM finds that entrepreneurship today is driven by enterprising individuals pursuing their ambitions while still employed elsewhere. Considering this trend, GEM finds that entrepreneurship has been on the rise since 2011.

Other polls are less sanguine. Speaking at the **December 2022 CSIS event**, panelist and Right to Start founder and CEO, Victor Hwang, noted that recently, “entrepreneurs [who were] asked ‘Is the system fair to you?’ answered with 81 percent saying the system is tilted and that the government favors large incumbent businesses instead of them. [Additionally] 69 percent of entrepreneurs say the system doesn’t care about them at all, and the data bears that out if you look at new entrants in government contracting.”

Meanwhile, the Bipartisan Policy Center **reported** in 2021 that the number of small businesses providing common products to the federal government has been declining from 2010 to 2019, with a 79 percent decline in new small business entrants to the federal procurements marketplace from 2005 to 2019—and with even less of this entrepreneurship coming from rural regions than in decades prior. A December 2020 CBO **report** points especially to a decline in the working-age population, particularly in rural areas, along with an unfavorable regulatory and financial environment for budding businesses. To address these issues, the report recommends changes to the visa system, adjustments to **noncompete** clauses, and better use of federal funding—some of which is now being tackled by the Biden administration as it works toward its stated **goals** of strengthening the U.S. workforce and equipping the middle class for success in the global economy.

Understanding these institutional rigidities provides an opportunity not only to increase overall entrepreneurial participation in the innovation ecosystem, but also to analyze where potential entrepreneurs are excluded. Creating a more flexible entrepreneurial ecosystem may also promote greater inclusion of diverse entrepreneurs with targeted, informed outreach efforts.

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Assessing Inclusivity in the Patent System

Diversity among entrepreneurs is one part of a complex picture. Innovators require a support system to become entrepreneurs, and given that many entrepreneurs are patent holders themselves, this holds true for the patenting ecosystem as well. Although downstream from the fundamental issues of better educational and professional opportunities that can foster entrepreneurialism in wider swaths of the U.S. population, diversity among inventors and patent holders is also essential to a robust innovation system. Patents assign property rights to research ideas that can become new products and services for the market. Well-defined intellectual property rights facilitate the sharing of new ideas, allowing inventors to collaborate with others to commercialize their ideas. But the set of patent holders is not diverse. According to a 2018 **study** in the *Quarterly Journal of Economics*, patent holders in the United States today are primarily white, male, and high income.

One source of this imbalance may be the underrepresentation of minority groups in the patent law industry—or even the legal profession at all. According to a 2020 United States Patent and Trademark Office (USPTO) **report**, *Diversity in Patent Law: A Data Analysis of Diversity in the Patent Practice by Technology Background and Region*, women make up 36 percent of all lawyers, and women of color only 9 percent of lawyers, and all racial minorities only roughly 17 percent. The report also documents the vast disparities in representation across all aspects of patenting, whether among USPTO attorneys (20 percent female and 5 percent racially diverse), or even among USPTO practitioner areas such as biotech, computer science, mechanical, etc., which all show nearly identical percentages of female and minority representation.

As former USPTO director, Andrei Iancu, noted at CSIS’s December 2022 event, “there are over 700,000 applications to the USPTO. The vast majority are represented by counsel, and I just wonder if counsel is fighting as hard for their women clients as they’re fighting for their male clients.” Although impossible to quantify, Almesha Campbell in response noted that countless would-be inventors and patent attorneys mention the problem of a lack of visible diverse representation among the patent law community. It is precisely overcoming this problem that will move the United States to have innovators that better reflect the diversity of the overall population.

Ensuring better representation of the U.S. population as patentors and patent experts must start from the bottom up. Recognizing this issue, some law **firms** like Harrity & Harrity, LLP have instituted a Minority Firm Incubator that seeks to “train, develop, and launch minority-owned patent law firms.” Yet, diversifying the composition of practitioners of patent legal expertise requires tackling the same pipeline issues that are central to the challenge of getting diverse and underrepresented individuals into the patent system.

Although it is true that the connection between university degrees in science, technology, engineering, and mathematics (STEM) and the frequency of patenting—and thus innovation—has been **proven**, often overlooked is the current practice that becoming a patent lawyer requires a hard science **degree**. In 2019, only 22.3 percent of bachelor’s **degrees** overall in the United States were in STEM fields, compared to 29-33

percent in nations like Germany, South Korea, and the United Kingdom. In addition, Pew Research **data** shows that the majority of the U.S. public considers the state of the nation's early STEM education lackluster compared to other countries. What is more, women, Black Americans, and Hispanics in the United States make up far less of the STEM degrees and workforce than their overall representation in the employed U.S. population should suggest, **according** to the National Science Board. Specifically, **according** to the National Center for Education Statistics, only 32.4 percent of U.S. STEM degrees awarded in the 2017-2018 academic year went to women—which, though low, is a 66.3 percent increase from the 2008-2009 school year of a decade before. A 2019 **study** by The University of Texas at Austin also found that, even though Black and Latino students declare their intent to major in STEM degrees at roughly the same rate as their white peers (at almost 20 percent), only 43 and 34 percent of those Black and Latino students, respectively, went on to attain that degree, compared to 58 percent of their white peers.

Ultimately, to improve the diversity of the patent law workforce, the United States should improve its education system by prioritizing and encouraging wider STEM education, and thus nurturing STEM talent. Although it is no small feat, tackling this challenge already shares overwhelming bipartisan consensus and has been the focus of **several** legislative efforts in Congress over the last few years alone.

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Along with greater educational investment in STEM and patent law to increase diversity, there needs to be a commercial system that supports and fosters the success of any action taken toward greater inclusion. Greater entrepreneurial support, private DEI efforts from large and small companies, capital for expansion, and technical assistance are all key to crossing the finish line and truly creating an inclusive innovation environment from the start to the end of the processes involved.

The Global View: Still a Ways to Go

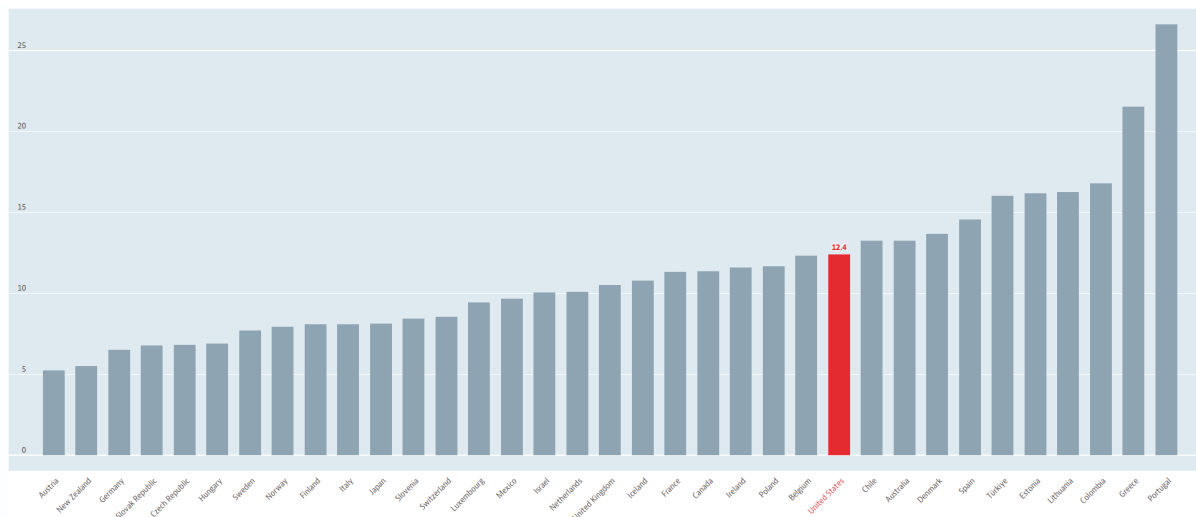
The lack of inclusion in the innovation system is not just a domestic issue, but a global issue. The issue manifests itself to varying degrees in other major developed G7 nations, as well. However, comparing data on inclusion across national innovation systems is a challenge because there is no standardization of the term “inclusion” across all countries due to different economic and cultural contexts.

Organization for Economic Cooperation and Development (OECD) data uses a common definition and metric of inclusion across these countries by measuring the participation of women, but looking at other factors like ethnicity and race is difficult due to different interpretations with regard to minority populations and cultural definitions of diversity in relation to race.

Although measurement remains a challenge, the potential gains from greater equity appear substantial. The 2015 McKinsey & Company **report**, *The Power of Parity: How Advancing Women's Equality Can Add \$12 Trillion in Global Growth*, posits that if countries modeled their gender parity goals on that of the fastest-improving country in their region, by 2025 the global GDP could grow by the equivalent

of Germany, Japan, and the United Kingdom’s GDPs combined. For much of the developing world, pursuing these goals begins with improving access to healthcare, education, and opportunities that many in the United States and other developed nations take for granted. Goal 5 of the UN **Sustainable Development Goals**, “achieve gender equality and empower all women and girls,” was developed in 2015 in pursuit of improving opportunities for women in all sectors of the economy by 2030. Yet, even among developed nations, the United States still has work to do to ensure greater female participation as inventors in the innovation economy.

Figure 1: Women, % of Total Inventors, 2019 or Latest Available



Source: Organization for Economic Cooperation and Development, *Entrepreneurship: Share of women inventors* (May 23, 2023), distributed by OECD.Stat, <https://stats.oecd.org/index.aspx?queryid=54676>.

Recommendations and Current Efforts

A culture that celebrates entrepreneurship—supported by a robust patent system—has made the United States the innovation leader that it is today. To succeed in a world that will compete on the basis of innovation, the United States needs to put the talents of more of its citizens to work.

The federal government recently allocated \$500 million in the **CHIPS and Science Act** to the EDA to expand the innovation ecosystem outside of existing innovation hubs by creating at least 20 hubs in areas that are not leading centers for technology. Carefully selecting these hubs and continuing to invest in these areas offers the promise of improving inclusion by potentially increasing investments into new entrepreneurial projects and expanding regional opportunities across the country. Further, if properly funded, CHIPS **authorizes** a total of \$2.23 billion over the next five years for manufacturing extension partnerships program offices (MEPs), which are a key part of the manufacturing ecosystem. The Hollings MEP program expansion under CHIPS and Science will be, at least in part, focused on targeted initiatives, specifically for entrepreneurship training and additional support for existing manufacturers and owners of color, as well as on the creation of new partnerships with historically Black colleges and universities. Likewise, the new Regional Innovation Engines **program** at the National Science Foundation (NSF)—through the NSF Directorate for Technology, Innovation and Partnerships—plans to support several regional

innovation ecosystems, spurring economic growth in regions that have not fully participated in the technology boom of recent decades.

Yet these programs are place-based programs, based on the “people prosperity vs place prosperity” **concept** introduced by the economist Louis Winnick in 1966. Advancements in knowledge about neighborhoods and individuals allow for the ability to both refine the concepts discussed by Winnick and measure individual development. Place-based programs have been shown to benefit places where resources are provided, but the benefits are not necessarily accruing to individuals most in need of support and economic empowerment. Improving measures to help federal agencies better understand who specifically is benefiting, rather than relying only on aggregate measures—even if at the neighborhood level—will improve program outcomes. This will take more time and effort, but the benefits of improving opportunities for populations with the greatest need will be better assured as a result.

Going forward, better measures of inclusion will help guide the implementation of policy. In this respect, current **efforts** call for updated reporting requirements and research support. There have also been proposals, such as the Individuals with Disabilities Education Act, that would permit the USPTO to voluntarily collect patent applicants’ demographic information for research and policy purposes. Implementing inclusion metrics like these could provide a more complete picture of current inclusion efforts, along with where and how improvements can be made. It is essential to use these metrics to instruct how to outline and implement community-informed efforts so that the individuals meant to benefit from regional hubs or greater access to the patent system are practically able and encouraged to participate.

To sustain and grow inclusion efforts over the long term, federal and state governments have been turning their attention to increasing educational opportunities. In 2019, Congress passed the **Building Blocks of STEM Act** to focus research efforts on STEM curriculums for elementary schools. Already, there **are six EDA programs** called Investing in America’s Communities, which focus on everything from pandemic relief funding to regional research grants, and even investment in Indigenous communities. Further, the **STEM Talent Challenge**, authorized in 2023 under the Stevenson-Wydler Technology Innovation Act of 1980, makes available \$4.5 million in grants to build regional STEM talent training systems.

In the meantime, some advocates have also called on the federal government to expand current affirmative action **requirements** for federal contractors to recruit and advance qualified minorities, women, and people with disabilities. Investments in regional hubs can also draw in sections of the U.S. population—those in the heartland—that have traditionally been left out of the prosperity generated by the innovation economy.

Although there is no panacea, the good news is that there is a growing recognition of the economic necessity for a more inclusive innovation economy, and this in turn is generating a growing debate on policies to support this outcome. In a rapidly changing global landscape, investing in a more inclusive innovation community is critical for the United States’ long-term economic success, social cohesion, and national security. ■

***Tom Guevara** is a senior adviser (non-resident) with the Renewing American Innovation Project at the Center for Strategic and International Studies (CSIS) in Washington, D.C. **Alexander Kersten** is deputy director and fellow with the CSIS Renewing American Innovation Project. **Srishti Khemka** was a research intern with the CSIS Renewing American Innovation Project.*

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